

AVANTIKA

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PROFESSIONAL SUMMARY

Data Scientist with hands-on experience building end-to-end machine learning systems in churn prediction, recommendation systems, and phishing detection. Skilled in Python, SQL, Machine Learning, NLP, Computer Vision, A/B Testing, Explainable AI, and dashboard development.

KEY SKILLS

Languages & Databases: Python, SQL, DuckDB

Data Extraction & Processing: PDF Data Extraction, Data Cleaning, Data Validation, Data Quality Assessment, Data Transformation, Data Standardization, Record Matching, Duplicate Detection

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, Tableau, Microsoft Excel

Machine Learning: Scikit-learn, XGBoost, Logistic Regression, Random Forest, Classification, Feature Engineering, Model Evaluation, Cross Validation, A/B Testing

Deep Learning & AI: PyTorch, Hugging Face Transformers, NLP, Computer Vision, Transfer Learning, EfficientNet, DistilBERT

Data Engineering & APIs: FastAPI, Data Cleaning, ETL Pipelines, Data Validation, CI/CD Pipelines, REST APIs

Analytics & Statistics: Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, Customer Segmentation, Market Basket Analysis, Treatment Effect Analysis

Business Intelligence & Reporting: Dashboard Development, KPI Reporting, Operational Analytics, Business Analytics, Performance Monitoring, Data Storytelling

AI & Explainability: PyTorch, Hugging Face Transformers, NLP, Computer Vision, SHAP, Grad-CAM

Tools & Platforms: Git, GitHub, Streamlit, Hugging Face Hub, Google Colab

PROJECTS

Q-Comm Market Basket Analysis

[GitHub](#)

- Built an end-to-end market basket analysis pipeline on 3.4M Instacart orders, simulating recommendation systems for quick-commerce platforms like Zepto and Blinkit through association-rule mining, customer segmentation, and product embeddings.
- Caught and fixed a hidden data-sampling bug that was inflating results, then shipped a FastAPI backend and a 4-page Power BI dashboard to track the corrected purchase patterns.

Indian Telecom Churn Analysis

[GitHub](#) | [Live Demo](#)

- Built an end-to-end churn prediction pipeline on 10,000 synthetic Indian telecom subscribers calibrated to real Jio/Airtel/Vi/BSNL market share, Selected Logistic Regression over Random Forest for superior churn recall on imbalanced data.
- Ran a 10% discount voucher A/B test (n=3,163) that cut churn by 45%, then used treatment-effect analysis to find one subscriber segment where the discount didn't work at all, saving ₹50K; deployed as an interactive Streamlit dashboard with live risk scoring and SHAP explanations.

PhQure — Phishing & Quishing Detection System (M.Sc. Dissertation)

[GitHub](#) | [Live Demo](#)

- Built a multi-modal phishing and QR code quishing detector combining three independent branches (URL features, QR images, raw text) fused through a meta-learner; deployed live with the fine-tuned model hosted on HuggingFace Hub.
- Individual branches hit up to 99.97% AUC, and the fusion layer automatically learned to downweight the one branch that was overfitting, validating the multi-branch design.

EDUCATION

Central University of Haryana, Mahendragarh, Haryana

July 2024-June 2026

Master of Science (Data Science) | CGPA: 8.10/10 | First Class with Distinction

Sophia Girls' College (Autonomous), Ajmer, Rajasthan

July 2019 – June 2022

Bachelor of Science (Mathematics) | CGPA: 8.92/10 (83.87%)

CERTIFICATIONS

Certified in Cybersecurity (CC) — (ISC)²

Aug 2025

- Gained knowledge in security principles, incident response, network security, and security operations.

ACTIVITIES

- Coding Club, Central University of Haryana — Committee Member
- SWAN (animal and nature society), Central University of Haryana — Committee Member
- Abdul Kalam Forum, Sophia Girls' College (Autonomous) — 3rd year representative